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Palaeomagnetism of nine dated Phanerozoic dykes in south-east Greenland

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Summary. Nine basic dykes were sampled near Angmagssalik, east Greenland. Specimens have been treated by alternating field demagnetization in 11 steps up to 3000 (peak) oersted (300 mT). The 'cleaned' direction at all sites is recognized after treatment at 150 oersted. All specimens are reversely magnetized. The mean of the site mean directions has declination = 182°.0, inclination = −66°.9, $k = 45$, $\alpha_{95} = 7°.7$. This direction yields a palaeomagnetic pole (reversed) at 73°.4 N, 139°.5 E ($dp = 10°.7$, $dm = 12°.9$) which is near, but significantly different from, that derived from lower Tertiary rocks in Greenland, namely 63°.2 N, 184°.6 E ($A_{95} = 4°.5$). K-Ar ages of the nine dykes, based upon whole-rock and mineral separates, range from mid-Tertiary to Cambrian. It is impossible to reconcile these ages with the palaeomagnetic results. The palaeomagnetic evidence, supported by geological inference, suggests that all nine dykes are members of the east Greenland lower Tertiary dyke swarm, designated THOL 1, of probable age *c.* 52 Ma.

The difference between the poles given above can be explained by supposing that the sampling area has tipped about a horizontal axis directed along 013°/193°, the angle of rotation being 13° (±11°) anti-clockwise, when the axis is viewed along 013°. This local effect could have been due to block faulting when the north-east Atlantic started to open, or may be attributed to upwarping of the coast due to the weight of the ice-cap inland.

1 Introduction

The dykes of this study were sampled in the course of palaeomagnetic work, mainly on the gneisses, during the Greenland Geological Survey's expedition to south-east Greenland in 1976. The main purpose of this expedition was the geological mapping of the early Precambrian rocks from Angmagssalik northwards. The Precambrian basement in this region is intruded by steep, basic dykes which were thought to be of Tertiary age (Bridgwater *et al.* 1977). Palaeomagnetic sampling of these dykes was undertaken to test this belief.

East Greenland is the site of intense early Tertiary igneous activity apparent in a number of igneous centres such as Skaergaard, and especially in the extensive flood basalts between

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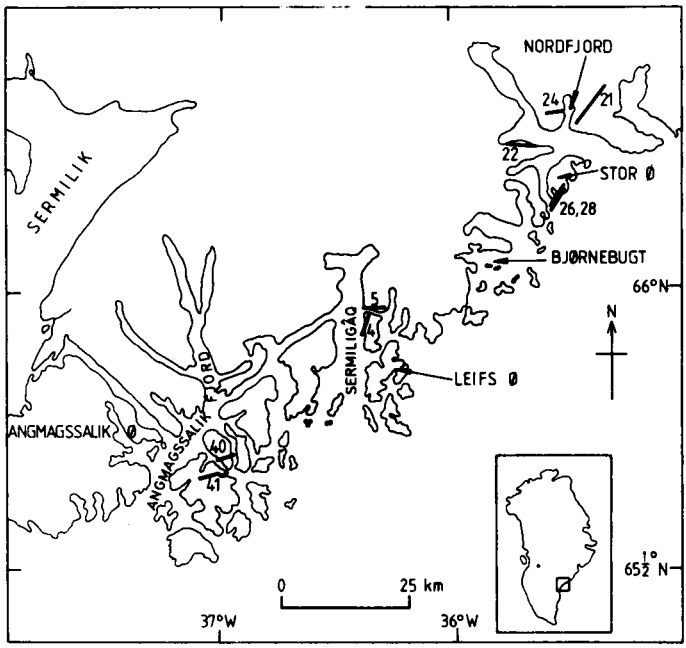


Figure 1. Map of the sampling area showing the sampled dyke locations and their strikes.

Kangerdlugssuaq and Scoresby Sund, 400–800 km to the north-east of the present area (Fig. 1). These basalts mark the initial rifting between central east Greenland and the Eurasian plate (Soper *et al.* 1976). They were probably all erupted in a single reversed epoch of the Earth's magnetic field within the time-range 52–55.5 Ma (Faller & Soper 1979).

The basalt pile near the coast dips seawards at 35°–55°. This is the manifestation of the east Greenland coastal flexure (Wager & Deer 1938). The basalts were intruded by a swarm of dykes, normal to the basalt surface, i.e. dipping to landward. Wager & Deer (1938) considered that the dykes were emplaced in that attitude as a consequence of the flexuring. On the other hand Nielsen (1975) suggested that they preceded the flexuring and were tipped with the basalts. An elegant palaeomagnetic test by Faller & Soper (1979) has discriminated in favour of the dykes being emplaced after the flexuring, which, these authors suggest, took place at 52–53 Ma. This is the main dyke swarm designated THOL 1 (Nielsen 1978), on the basis of petrology and geochemistry. Less extensive dykes, which are unambiguously post-flexural, were described by Nielsen (1978) and included in the palaeomagnetic study of Faller & Soper (1979).

The southern limit of the basalt is at Kangerdlugssuaq, apart from small patches, but it was initially more extensive. Much of the coast for 200 km south of Kangerdlugssuaq consists of basic dykes occupying more than 50 per cent of the outcrop (Bridgwater *et al.* 1978). However, in the area of the present study the dykes have died out to 1 or 2 per km. Most of them strike at 020°–040° but another group strikes at 060°–100°; both sets were included in the present study to test the possibility of an age difference between them.

2 Field work

Nine dykes were sampled, four of them with strikes 020°–040° and five in the range 071°–100°. The dykes are unmetamorphosed, fine-grained dolerites with brown weathering.

Samples were obtained by drilling. They were 2.5 cm diameter and typically 8 cm long. The number of samples per dyke varied between 7 and 15 but was usually 12. In only one case orientation was by sun-compass (Stone 1967). Lack of sunshine prevented the use of the sun-compass for the other dykes, which were orientated by Brunton compass in conjunction with topographic sighting. This method was proved good in this region by the use of both methods together at many of the gneiss sites. A typical value found for magnetic declination was 35° W.

The palaeomagnetic laboratory work was in two stages, the first at the Nuffield palaeomagnetic laboratory, University of Newcastle upon Tyne, and the second at ETH, Zürich.

3 Laboratory investigations, I

A specimen 2.5 cm long was cut from the lower end of each sample. The intensities and directions of magnetization were measured by a PAR SM-2 spinner magnetometer. Partial alternating field (AF) demagnetization was effected using a locally made machine, incorporating tumbling of single specimens about three axes simultaneously, in an ambient field of a few gammas (1 gamma = 10^{-6} mT).

All specimens were treated at AF fields of 100 and 150 peak oersted (Oe), (1 Oe = 10^{-1} mT). These treatments revealed stable magnetization in all specimens which was recognized by the following criteria:

(1) all sites have high values of the precision parameter, k (Fisher 1953);

(2) all specimens are reversely magnetized;

(3) the angular change in the magnetic vectors between the 100 and 150 Oe stages was small. For the site mean directions after these treatments the greatest angular change was 4°.5 (Table 1);

(4) eight of the nine dykes (all but site 26) form a cluster of palaeomagnetic poles (Fig. 2).

Nearly all palaeomagnetic studies on Tertiary rocks in Greenland and the Faeroe Islands, and the majority of those in the British Isles are characterized by reversed magnetization. Thus the common reversed polarity of the present dykes was in accordance with the expectation that they were Tertiary. At this point it was initially considered that the palaeomagnetic work was complete.

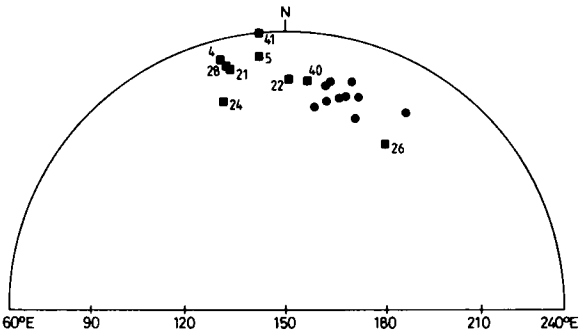


Figure 2. Equal area projection of the northern hemisphere displaying palaeomagnetic poles of the nine dykes (■), and palaeomagnetic poles from lower Tertiary rocks in Greenland from other studies (●) (Table 2).

Table 1. Palaeomagnetic data.

	Magnetization after 150 Oe* (final results)										Some mean vectors subtracted				
Site	*N*	Dec	Inc	*k*	α_{95}	ANG	Palaeomagnetic pole (reversed) °N	°E	*dp*	*dm*	*N*	Step (Oe)*	Dec	Inc	*k*	α_{95}
4	15	202.6	−70.1	368	2.0	0.4	73.7	90.0	3.0	3.4	13	350–550	216.1	−73.3	61	5.4
5	13	188.3	−71.4	526	1.8	1.0	79.2	117.9	2.8	3.2	11	350–550	197.0	−72.2	384	2.3
21	7	197.7	−67.9	838	2.1	0.2	72.2	105.7	2.9	3.5	5	250–350	193.4	−70.2	312	4.3
22	12	176.5	−65.4	936	1.4	0.8	71.1	151.7	1.9	2.3	10	350–550	177.1	−61.6	109	4.7
24	12	197.1	−58.3	54	6.0	4.1	61.0	116.4	6.5	8.8	12	100–150	195.5	−66.2	33	7.7
26	12	146.1	−46.7	231	2.9	4.5	46.5	190.1	2.4	3.7	10	250–350	145.4	−46.0	18	11.8
28	12	199.3	−68.1	571	1.8	2.0	72.1	101.9	2.6	3.1	10	350–550	192.6	−68.2	683	1.9
40	12	167.9	−65.4	962	1.4	0.4	70.7	168.4	1.8	2.3	10	550–750	164.4	−67.4	143	4.1
41	10	192.3	−76.5	4192	0.7	0.5	84.7	61.0	1.3	1.4	8	350–550	190.3	−77.4	364	2.9
Mean	9 (sites)	182.0	−66.9	45	7.7		73.4	139.5	10.7	12.9						

Notes:

N = number of specimens (one specimen per sample).

k and α_{95} are the precision parameter and the semi-angle (in degrees) of the cone of 95 per cent confidence about the mean direction (Fisher 1953).

ANG is the angle (in degrees) between the mean directions after treatment at 100 and 150 Oe.

dp and *dm* are the semi-axes of the ovals of 95 per cent confidence about the palaeomagnetic pole, *dp* in the colatitude direction and *dm* at right angles to *dp*.

Mean site coordinates: 66°.1 N, 36°.0 W.

* Values in mT are obtained by dividing tabulated values by 10.

Table 2. Lower Tertiary palaeomagnetic poles from other studies.

Rocks	Region	Palaeomagnetic poles °N	°E	Authors	Notes
Basaltic lavas	Ubekendt I., West Greenland	54	208	Tarling & Otulana (1972)	
Basaltic lavas	Disco I., West Greenland	67.5	195	Athavale & Sharma (1975)	
Basaltic lavas	Disco I., West Greenland	62	191	Deutsch & Kristjansson (1974)	Described as late-Cretaceous–Tertiary; one flow has a K-Ar age of 70 ± 4 Ma
Basaltic lavas	Scoresby Sund, East Greenland	69.7	181.4	Hailwood (1977)	
Basaltic lavas	Scoresby Sund, East Greenland	63	174	Tarling (1967)	
Basaltic lavas	Kangerdlugssuaq	63.4	185.1	Faller (1975)	
THOL 1 dykes	Kap Irminger, East Greenland	62.9	180.4	Faller & Soper (1979)	
Basaltic lavas	Jacobsen Fjord, East Greenland	55.8	183.0	Faller & Soper (1979)	Palaeomagnetic poles calculated by the present author and published here by kind permission
Post flexure dykes	Jacobsen Fjord, East Greenland	67.8	177.7	Faller & Soper (1979)	
Skaergaard gabbroic intrusion	Skaergaard, East Greenland	61	165	Schwarz, Coleman & Cattroll (1979)	The magnetization was probably established immediately after the occurrence of the East Greenland coastal flexure. Brooks & Gleadow (1977) dated emplacement at 64.5 ± 1 Ma via fission-track in zircon

Mean palaeomagnetic pole from the 10 studies = 63°.2 N, 184°.6 E, $k = 114$, $A_{95} = 4°.5$, where A_{95} is the semi-angle of the cone of 95 per cent confidence about the mean pole, assuming a Fisherian distribution of poles.

4 Consideration of radiometric ages

Comparison of the palaeomagnetic poles of these dykes with other Tertiary poles for Greenland (Table 2) indicated that the dyke poles (reversed) tend to have more westerly values of longitude (Fig. 2). This observation rendered radiometric dating desirable. Two specimens from each dyke were sacrificed for this purpose, and a K-Ar study of these carried out (Mitchell 1978). In four dykes whole rock ages were obtained and in five cases both whole rock ages and ages from mineral separates were found. A remarkable spread of ages emerged, four Tertiary, three Cretaceous, one Jurassic and one Cambrian (Table 3). These dates were extremely difficult to reconcile with the palaeomagnetic results. For example the Cretaceous and Jurassic eras were characterized by predominantly normal polarity. Nevertheless, apart from polarity, it was possible to explain the palaeomagnetic poles for the 'Cretaceous' and 'Jurassic' dykes (sites 28, 26, 21, 24, Table 1) because a survey of the literature showed that there was no distinct movement of the palaeomagnetic pole for Greenland between late Triassic and mid-Tertiary times. However, the inclusion of a Cambrian pole in a Tertiary group could not be justified; the palaeomagnetic pole from site 22 with the Cambrian age is remote from palaeomagnetic poles from Cambrian kimberlite dykes in west Greenland (Beckmann & Mitchell 1976).

One approach to the paradox would be to postulate that some of the dykes were remagnetized in Tertiary times by the basalts, whereas the older radiometric ages were not reset. To test this possibility a very detailed AF demagnetization programme was carried out.

5 Laboratory investigations, II

At the palaeomagnetic laboratory of ETH, Zürich, a locally made AF demagnetizer was available, capable of holding six specimens in the tumbler simultaneously and attaining a demagnetizing field of 3000 Oe, which is exceptionally high. All the remaining specimens

Table 3. K-Ar data for the nine dykes (Mitchell 1978).

Site no.	Material	Age (Ma)	Strike
41	Whole rock	45.8 ± 1.5	080°
5	Whole rock	54.9 ± 1.4	090°
4	Whole rock	58.6 ± 1.5	020°
	Plagioclase	62.1 ± 1.5	
	Pyroxene	62.0 ± 2.0	
40	Whole rock	61.9 ± 1.6	071°
28	Whole rock	78.5 ± 1.5	034°
	Plagioclase	75.6 ± 1.8	
	Pyroxene	76.7 ± 2.2	
26	Whole rock	83.0 ± 1.8	034°
21	Whole rock	102 ± 2	040°
	Whole rock	105 ± 2	
	Plagioclase	82.2 ± 1.5	
	Pyroxene	92.9 ± 2.0	
24	Whole rock	164 ± 3	085°
	Plagioclase	159 ± 3	
	Pyroxene	178 ± 3	
22	Whole rock	548 ± 6	100°
	Plagioclase	552 ± 6	
	Pyroxene	558 ± 8	

were demagnetized at the following alternating fields: 250, 350, 550, 750, 900, 1200, 1800, 2400, 3000 Oe, within an ambient field of a few gammas. Intensities and directions of magnetization were measured on a Digico balanced fluxgate spinner magnetometer (Molyneux 1971).

6 Data analyses

For most specimens, directions of magnetization remained remarkably steady during demagnetization through several demagnetizing fields, whilst values of k remained large. Even after treatment at 900 Oe, $k = 265$ for site 40, and after 1200 Oe, $k = 42$. In site 22 (the 'Cambrian' dyke) after 1200 Oe, $k = 38$. In both cases there is very little angular movement of the magnetic vector from earlier stages. The magnetization resides in magnetite and the high coercivity can be ascribed to single domain grains (Evans & McElhinny 1969), so that the directions of magnetization in these dykes are particularly reliable.

For demagnetizing fields $\leqslant 900$ Oe angular movements are often so small that mean directions cannot be plotted as distinct points on stereograms. For this reason the vectors have been represented (Fig. 3) in Zijderveld diagrams (Zijderveld 1967). This technique is particularly valuable for detecting multi-component magnetization. A change in direction in these plots implies a change in the component being removed during demagnetization. A straight line directed towards the origin means demagnetization of a single component.

Usually Zijderveld plots show the behaviour of single specimens. However, in the present case plots which demonstrate the site-average behaviour of the specimens have been preferred; the unit vector associated with the site mean direction after each demagnetizing step was resolved into north, east and down components which were then multiplied by the mean intensity of magnetization remaining, divided by the mean initial magnetization. Each point is associated with uncertainty, which increases as the origin is approached. All those mean directions have been plotted which are significant at the 95 per cent confidence level. In some of the plots, such as site 28, initial sections indicate removal of components arising from the Earth's present field; thereafter almost straight sections directed towards the origin indicate the removal of the main magnetization. In the case of site 41 there is no secondary component of magnetization.

In all cases except site 26 the conclusion from study of the Zijderveld diagrams is that from about 150 Oe onwards a single component of magnetization is being removed up to high demagnetizing fields. (It may be recalled that site 26 has an anomalous palaeomagnetic pole. At this site $k \leqslant 2.8$ for demagnetizing fields of $\geqslant 350$ Oe and so it is not good palaeomagnetic material.)

This conclusion has been justified by considering further the mean vectors removed after each demagnetizing step. All nine sites are initially reversely magnetized. At six of them the mean vector removed in the 0–100 Oe step is normal whereas at the remaining three this vector is reversed. Examples of the two types are sites 22 and 21, respectively. These are illustrated by the variation of the precision parameter, k, with demagnetizing field for the total vector and the vector subtracted at each stage (Fig. 4). The original assignment of cleaned directions after 150 Oe appears to have been a good choice because in every case the mean direction after 150 Oe is very close to that found for mean vectors subtracted in the 100–150 Oe step or a later step (Table 1).

As in all palaeomagnetic work, it was not possible to reduce the intensity of magnetization to zero, on account of noise from the demagnetized grains. However, at five of the sites the mean direction is significant, at the 95 per cent confidence level, even after treatment at 3000 Oe. Detailed statistical investigations and displacements seen in the Zijderveld diagrams

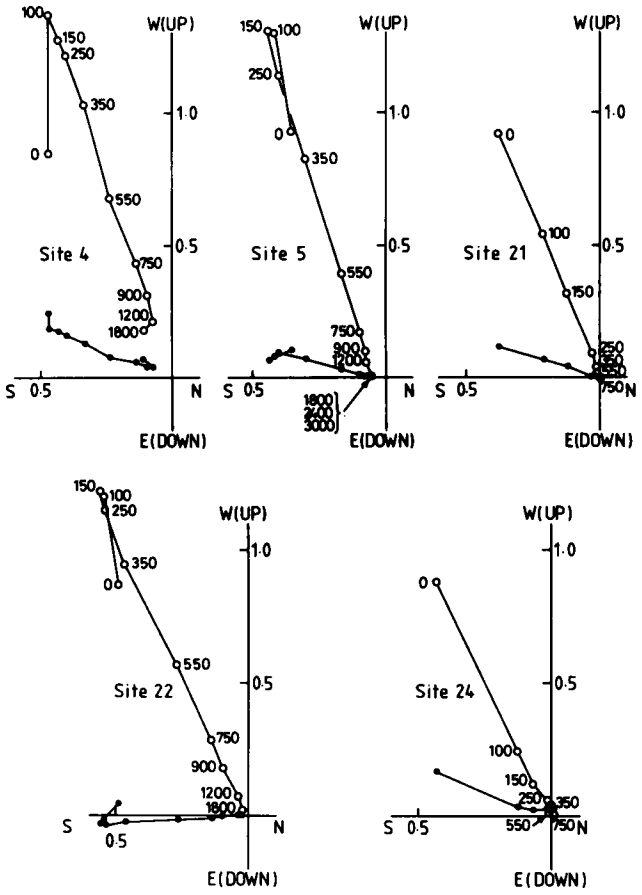


Figure 3. Orthogonal demagnetization projections (Zijderveld 1967) of the successive end-points of site mean magnetization vectors during progressive demagnetization (see text for intensity normalizing procedure). Solid circles represent projections on to the horizontal plane, and open symbols those on the north–south vertical plane. Demagnetizing fields in Oe (10^{-1} mT).

indicate that small high coercivity components remain after high alternating fields. The corresponding poles have low latitudes. A similar effect was found by Faller & Soper (1979) using thermal demagnetization near the Curie point of magnetite. In the present case, these residual site mean directions formed a random set. Perhaps anhysteric or rotational remanent magnetizations were responsible for the residual magnetizations, in spite of attempts to minimize them by mounting the six specimens as randomly as possible in the tumbler. However, systematic mean directional movements between demagnetizing steps render this possibility unlikely. Although the effect is not understood it is considered irrelevant to the main theme of this paper.

Another method of selecting the best cleaned direction for each site was tried but rejected. For each specimen the magnetic stability index (Briden 1972) was calculated for each demagnetizing stage. The stage for which this index was a maximum was picked out and the site mean direction was formed by combining such directions. For most sites this procedure worked well but it was extremely prejudicial to site 24. At this site demagnetization caused the intensity of magnetization to fall rapidly (Fig. 3); after 150 Oe, the mean intensity was only 14 per cent of the initial intensity. These low demagnetizing stages gave

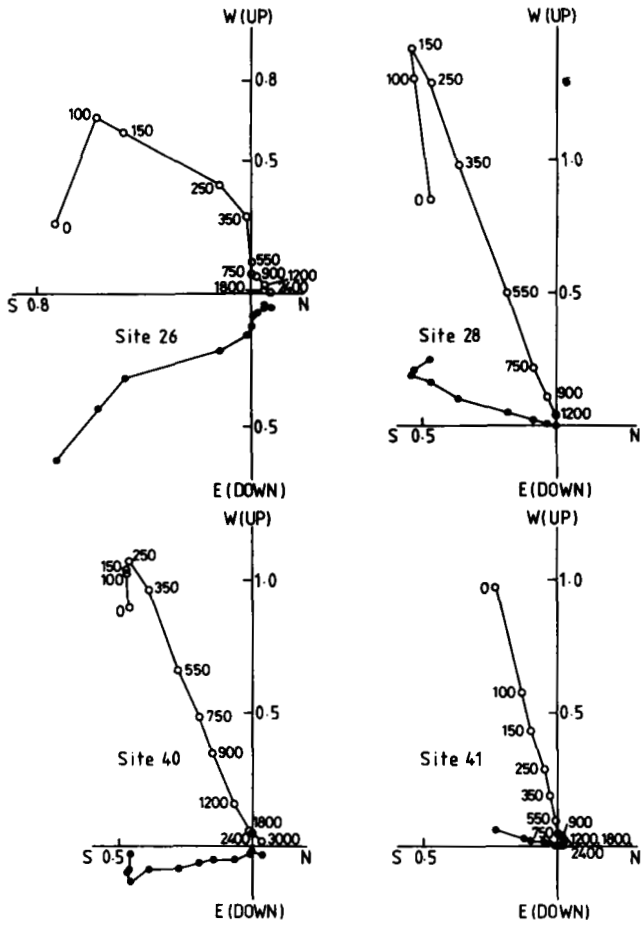


Figure 3 – *continued*

rise to low values of the stability index, which takes account of changes in intensity as well as direction. The highest values of the stability index for site 24 were found within scattered groups resulting from demagnetization at higher fields. Fortunately it was possible to prove by a contact test that the direction of magnetization after 150 Oe (or 100 Oe) represented the direction established when this dyke was emplaced.

7 Contact tests

Site 24 is a dyke 1.2 m wide and cuts a pseudotachylite at 55°. Palaeomagnetic data from the pseudotachylite showed that it has an early Proterozoic age (Beckmann 1977). Ten samples were drilled in the pseudotachylite at the following perpendicular distances from the nearer margin of the dyke:

Sample 1 : 0.3 m
2–5 : 0.9 m
6 : 3.0 m
7 : 3.7 m
8–9 : 3.8 m
10 : 4.3 m.

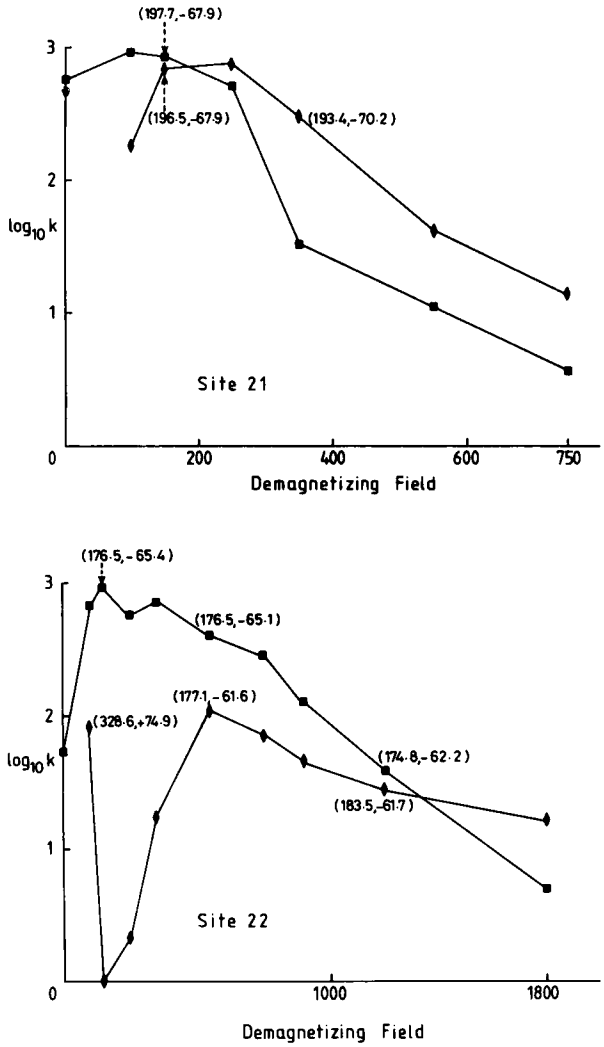


Figure 4. Graphs of $\log_{10} k$ versus demagnetizing field (unit = 1 Oe = 10^{-1} mT) for total vectors (■) and vectors subtracted (♦) for sites 21 and 22. Numbers in brackets are directions of magnetization.

Specimens were treated at 50 and 100 Oe and then thermally demagnetized at 500, 540, 560 and 580°C. In specimens 6 to 10 the effect of the dyke cannot be detected; the original magnetization of the pseudotachylite is normal (downwards) and is most clearly seen in specimens 7–10 after 540°C (Fig. 5). Specimen 1, closest to the dyke, has been strongly affected by it; the components removed in three demagnetizing steps fall close to the dyke direction (Fig. 5). Fig. 5 also shows that specimens 2 to 5 were partially remagnetized by the dyke.

Palaeomagnetic work on the gneisses constitutes a less specific contact test. Gneisses sampled approximately 200, 70 and 150 m from dyke sites 21, 22 and 28, respectively, display directions of magnetization which are of early Proterozoic age (the 'Nordfjord gneisses' in table 1, Beckmann 1979).

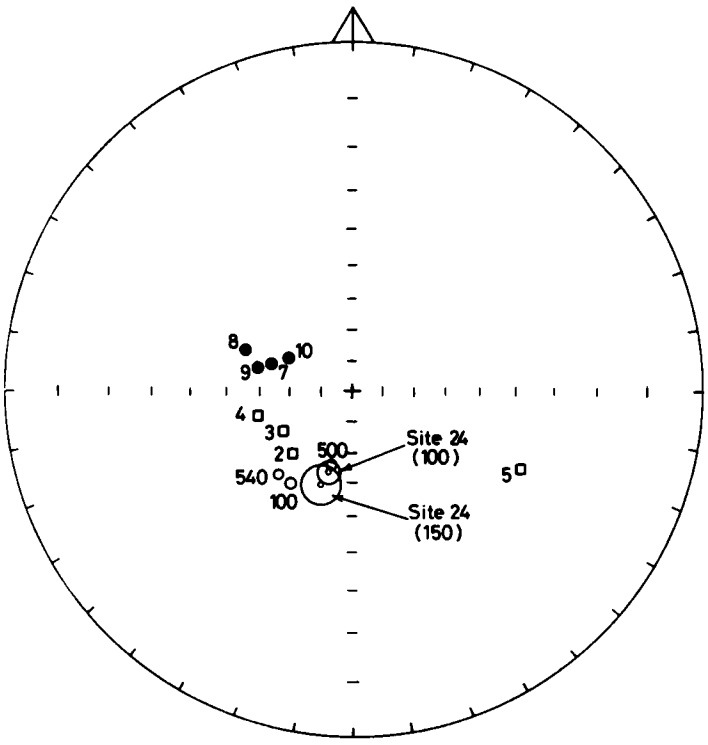


Figure 5. Stereogram showing the mean directions of magnetization of site 24, surrounded by circles of 95 per cent confidence, after 100 and 150 Oe. Also shown are directions of magnetization for pseudotachylite (PT) specimens 7 to 10, remote from the dyke, after treatment at 540°C. The points 100, 500 and 540 represent the components *removed* from PT specimen 1 in the stages 50–100 Oe, 100 Oe–500°C, and 500–540°C, respectively. Points 2 to 5 represent components *removed* in PT specimens 2–5 in stages 50–100 Oe (2, 4, 5) and 100 Oe–500°C (specimen 3). Open (closed) symbols denote reversed (normal) magnetization.

8 Conclusions

These contact tests refute the hypothesis of regional remagnetization and it is concluded that the cleaned directions of magnetization in the dykes were established at the times of their emplacement into cool rocks.

The close grouping of the palaeomagnetic poles (apart from site 26) implies similarity of age. This implication is reinforced by their common reversed polarity, suggesting intrusion in one or two reversed epochs of the Earth's magnetic field. The dykes are probably part of the main dolerite swarm of east Greenland, of THOL 1 type, emplaced at 52–53 Ma (N. J. Soper 1980, private communication).

No effort has been spared to reconcile the K-Ar ages (Mitchell 1978) with the palaeomagnetic results but this has turned out to be impossible. Indeed there is a widely held belief that in east and north Greenland excess argon in the Tertiary rocks is a serious problem, sometimes even resulting in Precambrian 'ages'. In north Greenland 'ages' of dolerite dykes sometimes exceed those of the enclosing rocks (N. J. Soper 1980, private communication).

The anomalous palaeomagnetic pole for site 26 does not imply an age difference from the other dykes because it lies within 100 m of site 28 and has the same strike; in the field it was obvious that it was a member of the same swarm.

The mean palaeomagnetic pole from the nine dykes is 73°.4 N, 139°.5 E, whereas that from the other studies (Table 2) is 63°.2 N, 184°.6 E. Can this difference be explained? The first possibility is that the difference is associated with the east Greenland coastal flexure. However, an attempt to correct the poles of the present study on this account would have *increased* the difference noted above. In any case Faller & Soper (1979) showed that the THOL 1 dykes were magnetized after the flexuring.

Evidently the only way of reconciling the two mean poles is to postulate that some local effect caused the difference. Let us suppose that the lower Tertiary palaeomagnetic pole in this region originally coincided with the pole from the rest of Greenland but that some tectonic rotation produced the observed difference. The calculation is more readily carried out by considering directions of magnetization than palaeomagnetic poles. If the palaeomagnetic pole were at 63°.2 N, 184°.6 N ($A_{95} = 4°.5$), the direction of magnetization *to be expected* in the region of study (66°.1 N, 36°.0 W) is dec = 336°.5, inc = 61°.5 ($\alpha_{95} = 3°.8$). The *observed* direction is: dec = 2°.0, inc = 66°.9 ($\alpha_{95} = 7°.7$). The two directions are significantly different because their cones of 95 per cent confidence do not intersect. One can account for this difference by supposing that the region of study has rotated 13° (±11°) anticlockwise about a horizontal line striking 013°/193° when the line is viewed along the direction 013°, the error being expressed at 95 per cent confidence limits. For the two mean directions and a horizontal axis of rotation this solution is unique, i.e. we could not have an axis with a different strike associated with a different angle of rotation. Unfortunately there are no Cambrian to Tertiary sedimentary formations in the area which could provide palaeohorizons, and thus allow the theory to be tested.

An explanation of such tipping may be sought in the separation of Greenland from Eurasia, one of whose effects might have been rotational block faulting (Cloos 1928) arising from the complex tectonics of rifting.

Alternatively the proposed tipping might be up-warping of the coast due to the weight of the ice-cap, which is manifested by the mountains at the coast and the nunataks inland. Mount Forel, 3360 m, the highest mountain in Greenland, is a good example, lying only 100 km NNW of the region of study.

The axis of the proposed tipping is parallel to Sermilik and Angmagssalik fjords (Fig. 1), which have deep extensions, as much as 965 m deep (Larsen 1975); these may be structural expressions of the tipping detected palaeomagnetically.

It is concluded that some such local movement of the coast may have occurred in the area of study but was absent in other parts of Greenland, in particular the Scoresby Sund and Kangerdlugssuaq regions of east Greenland.

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